Architecture 1001: x86-64 Assembly ThereWillBe0xb100d.c

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OST PRESENTS



ThereWillBe0xb100d.c

```c
int main(){
    char buf[40];
    buf[39] = 42;
    return 0xb100d;
}
```

ThereWillBe0xb100d.c

```
main:
0000000140001010  push          rdi
0000000140001012  sub           rsp,60h
0000000140001016  mov           rdi,rsp
0000000140001019  mov           ecx,18h
000000014000101E  mov           eax,0CCCCCCCCh
0000000140001023  rep stos      dword ptr [rdi]
0000000140001025  mov           eax,1
000000014000102A  imul          rax,rax,27h
000000014000102E  mov           byte ptr [rsp+rax+28h],2Ah
0000000140001033  mov           eax,0xb100d
0000000140001038  mov           edi,eax
000000014000103A  mov           rcx,rsp
000000014000103D  lea           rdx,[__xi_z+1A0h (0140006910h)]
0000000140001044  call          _RTC_CheckStackVars (01400010B0h)
0000000140001049  mov           eax,edi
000000014000104B  add           rsp,60h
000000014000104F  pop           rdi
0000000140001050  ret
```



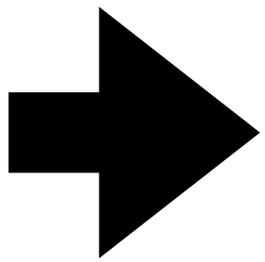
STOP
&&
STEP THROUGH
THE ASSEMBLY
&&
DRAW A STACK DIAGRAM



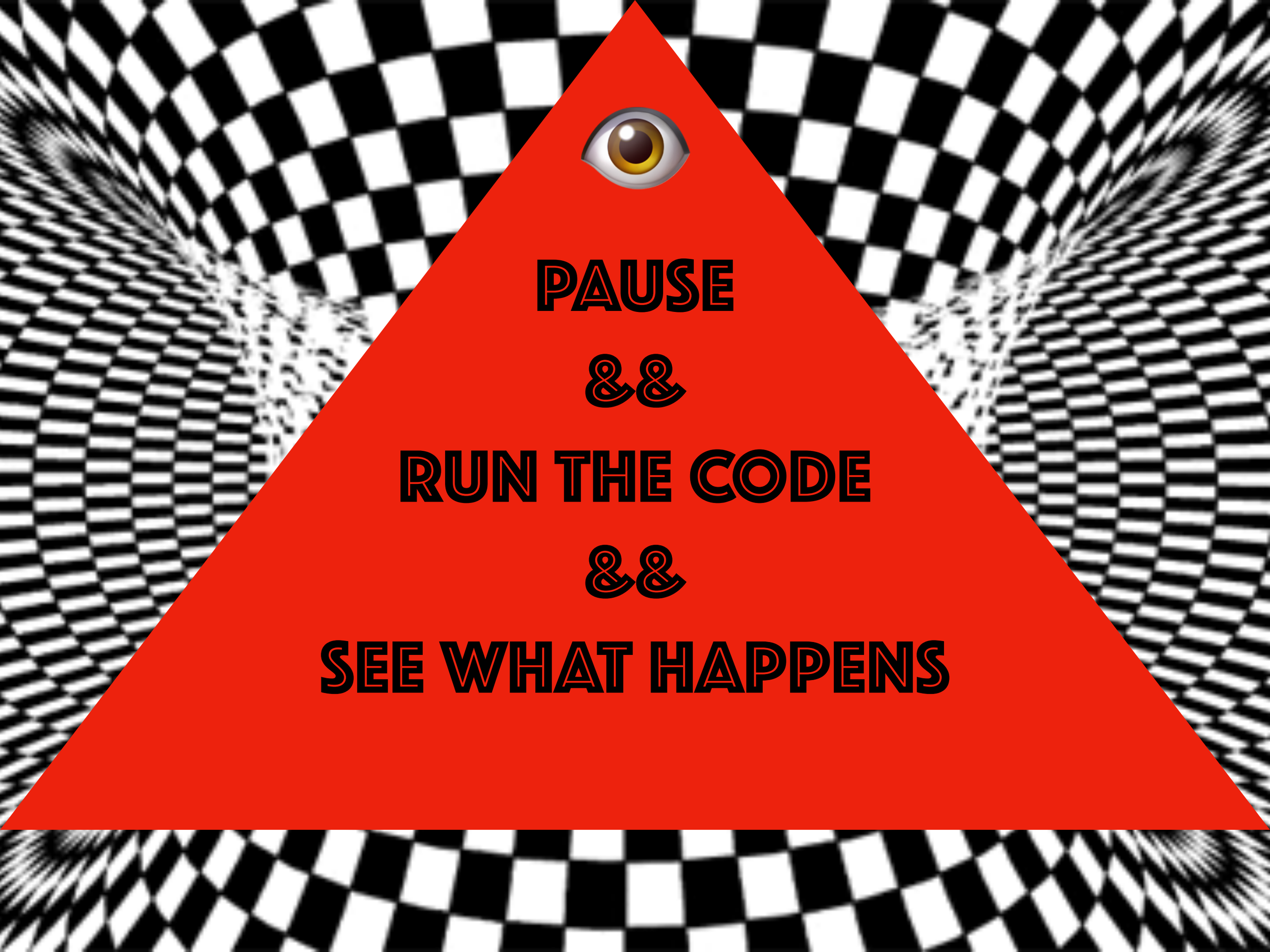
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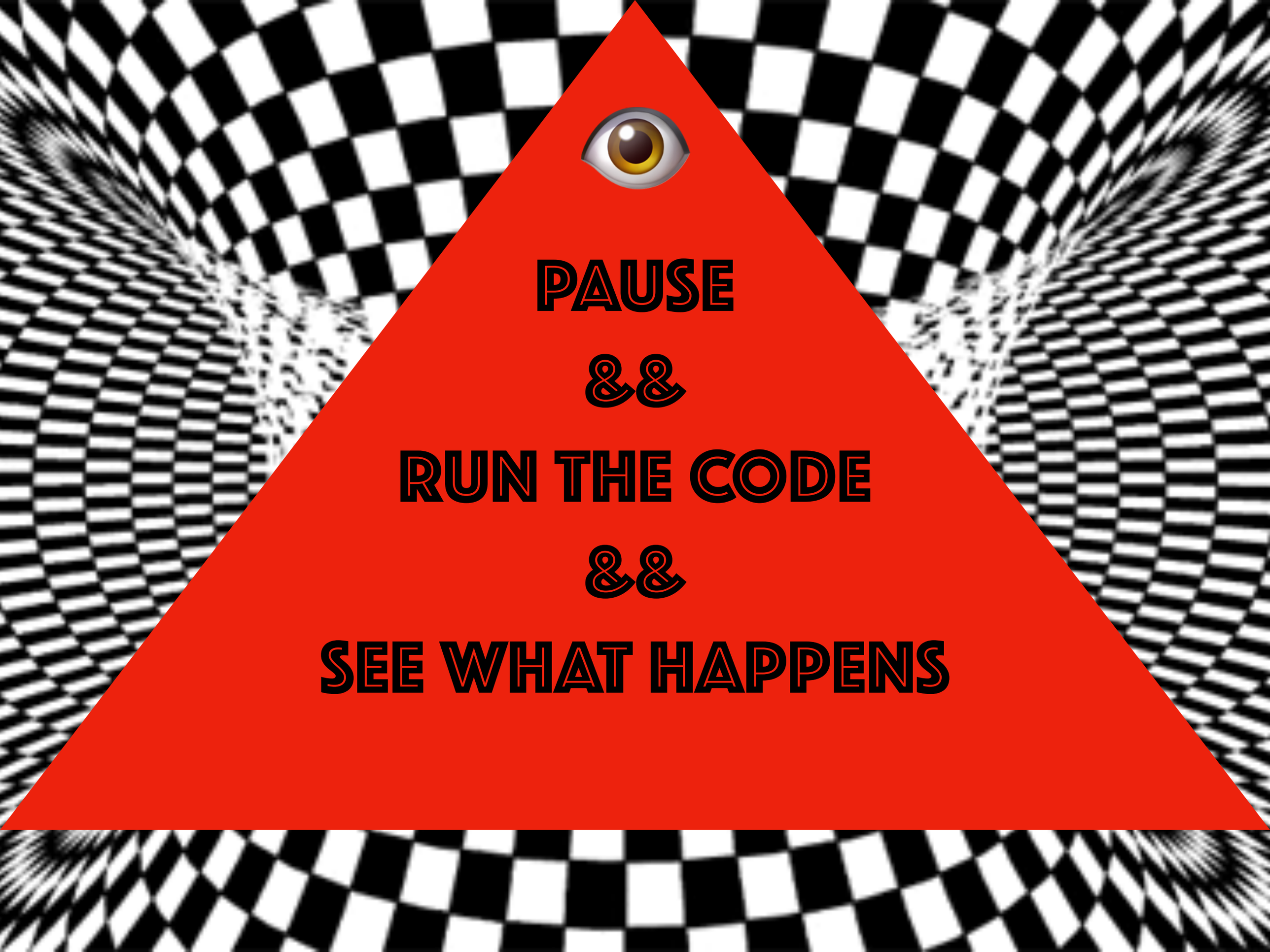
ThereWillBe0xb100d.c

Try writing outside the bounds of the buffer

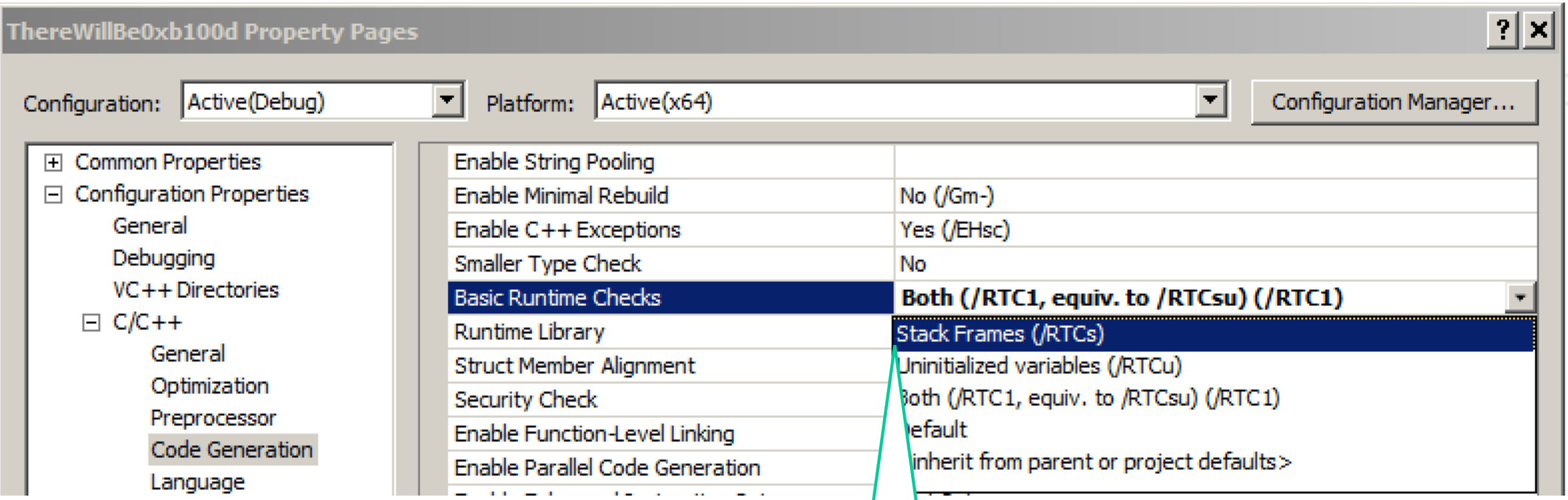


```c
int main(){
    char buf[40];
    buf[41] = 42;
    return 0xb100d;
}
```


PAUSE
&&
RUN THE CODE
&&
SEE WHAT HAPPENS


PAUSE
&&
RUN THE CODE
&&
SEE WHAT HAPPENS

Q: Where does the rep stos come from in this example?



A: Compiler-auto-generated code. From the stack frames runtime check option. This is enabled by default in the debug build. Disabling this option removes the compiler-generated code.

More straightforward without the runtime check

```
main:
0000000140001010  sub  rsp,38h
0000000140001014  mov  eax,1
0000000140001019  imul rax,rax,27h
000000014000101D  mov  byte ptr [rsp+rax],2Ah
0000000140001021  mov  eax,0B100Dh
0000000140001026  add  rsp,38h
000000014000102A  ret
```